

Project Portfolio

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Machine Vision

Deep Learning

Summer Work 2026

GreenAlVision

SAMK RoBoAI · Boliden collaboration

PROJECT OVERVIEW

Real-time AI system for Boliden's WEEE recycling conveyor belt. Semantic segmentation classifies every pixel into background, copper-rich copper-rich cable waste, or shredded PCB fragments — enabling data-driven data-driven purity measurement replacing manual estimation.

WHAT I BUILT

Rebuilt the full pipeline from a 65% accuracy prototype: switched to SegFormer B2 transformer with pretrained weights, built a line camera data pipeline with NIR channel integration, adapted the model to 4-channel inference, and built a retraining UI for flagging new materials.

IMPACT

Transforms chaotic scrap into a real-time purity metric — data-driven furnace decisions replacing manual estimation.

STACK

PyTorch · SegFormer B2 · ONNX · FastAPI · WebSocket · Spinnaker SDK ·
Roboflow

Role: AI/ML Engineer · Model development, dataset pipeline, line camera integration, NIR channel, retraining pipeline

92.1%

mIoU accuracy

+27pp

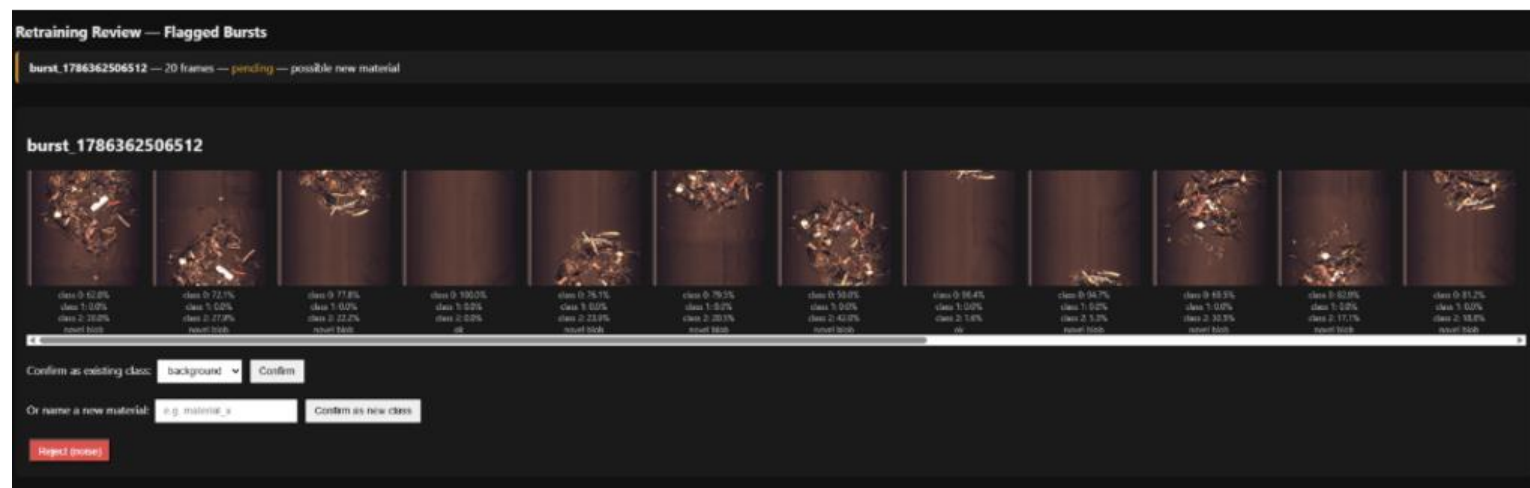
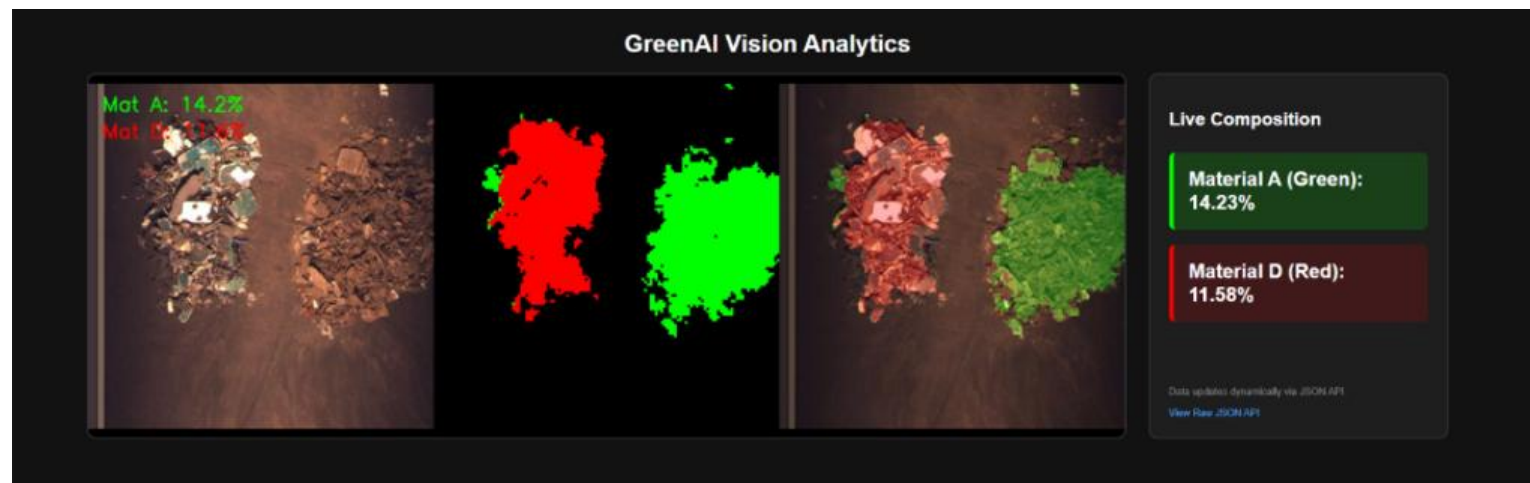
vs baseline

4-ch

RGB + NIR

<43ms

per frame



AI Handling of Land Use Documents

RoboAI Academy

PROJECT OVERVIEW

Designed and piloted an AI-supported workflow for reviewing land-use plans within a regional planning authority. Analyzes plan documents and spatial data to assess compliance with regional plans, planning objectives, and legal requirements.

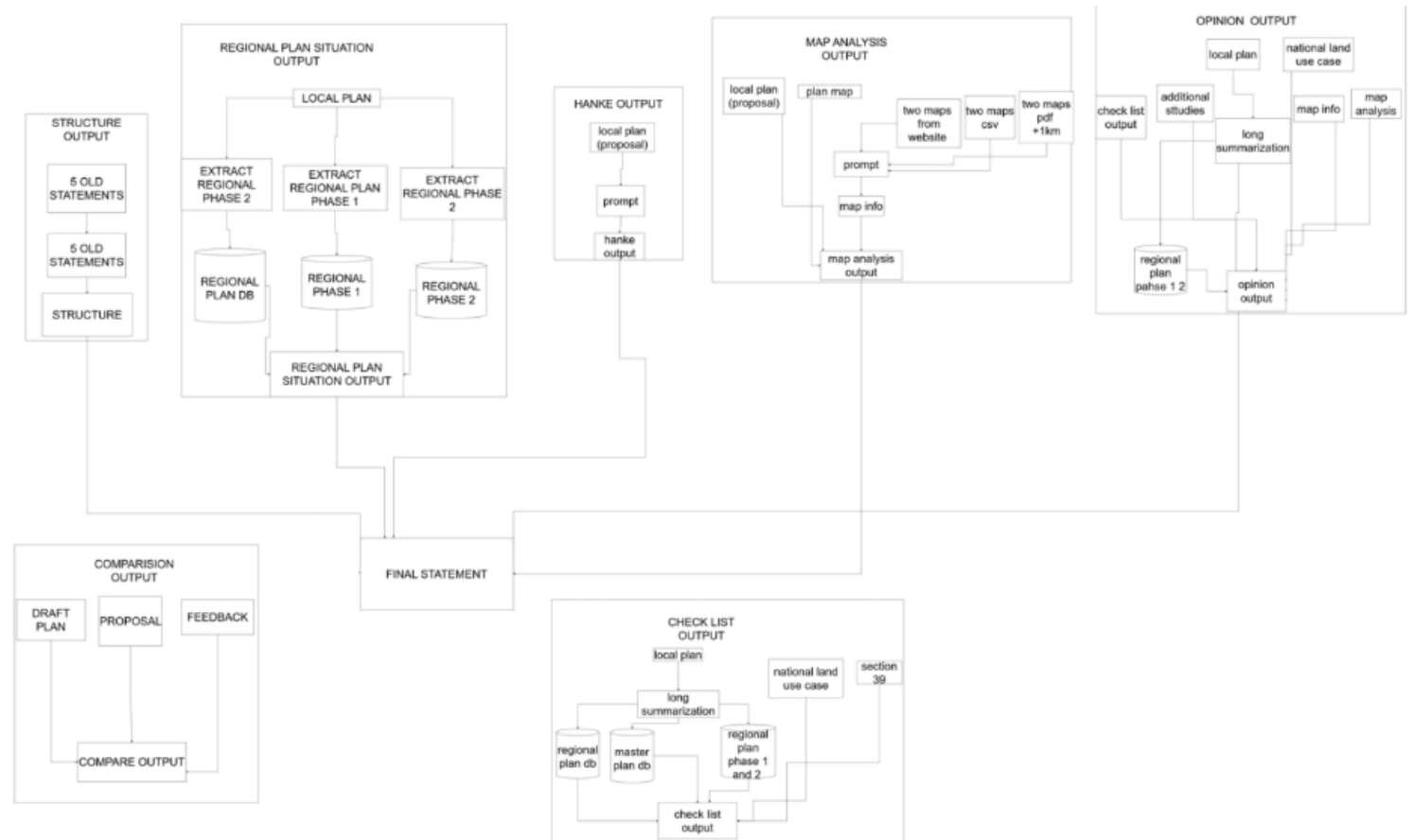
TECHNICAL IMPLEMENTATION

Developed checklist-based evaluation logic and map overlay analysis to support expert statements, identify planning conflicts, and detect regulatory gaps. Runs fully locally in n8n.

SKILLS

AI Decision Support · Spatial Data Analysis · Urban & Regional Planning · Land Use Law (Finland) · GIS & Process Design

Role: Communication Manager · Client communication, prompt engineering, input engineering, input validation and output evaluation



Machine Vision

Quality Inspection Demo

RoboAI Academy

SYSTEM DEVELOPMENT

Machine vision system for industrial faucet quality inspection. Detects missing markings, surface defects, and line deviations at production line speed.

PLATFORM EVALUATION

Evaluated Cognex, HALCON, and OpenCV. Selected and implemented the Cognex solution based on accuracy and integration requirements.

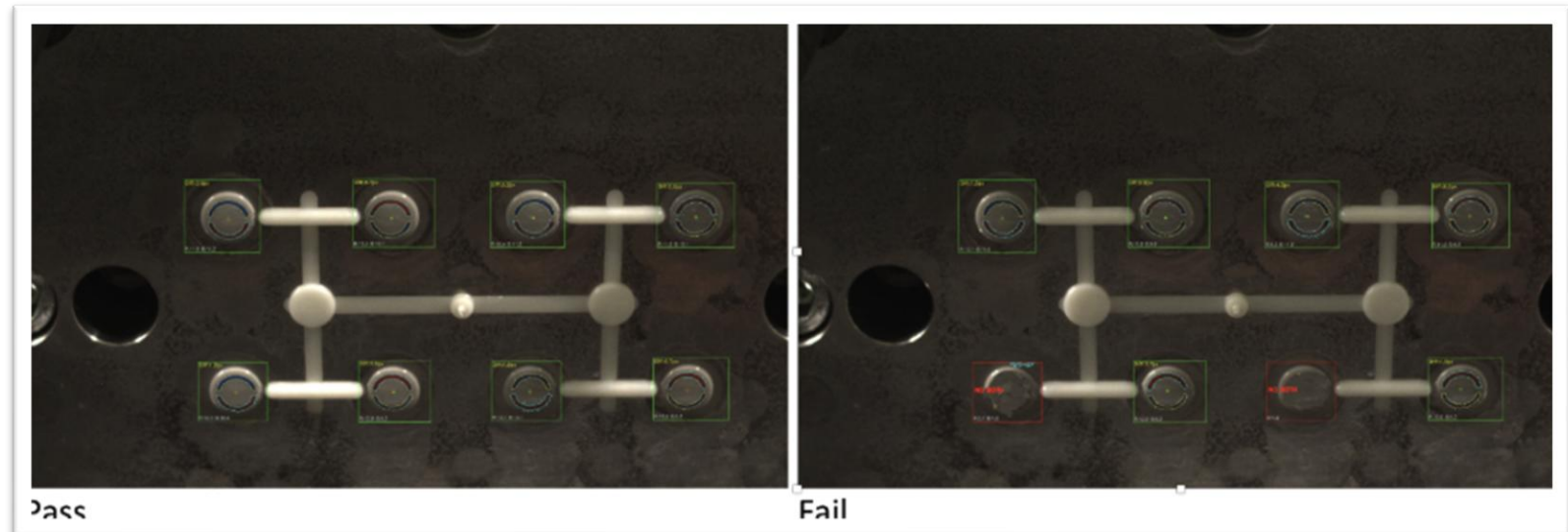
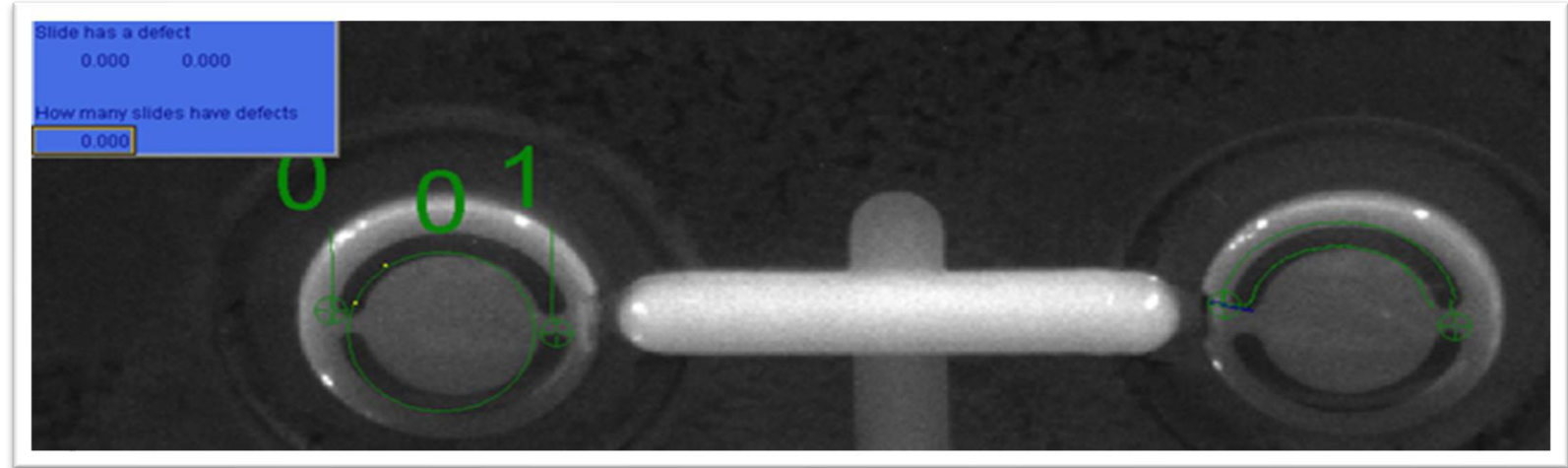
PIPELINE

Image capture → Defect detection → Automated pass/fail → Data logging.

SKILLS

Machine Vision · Cognex · Image Processing · Quality Control · Industrial Automation

Role: Project Manager · Team management, Cognex solution development, presenting results to client



SimpleFlow

RoboAI Academy

PROJECT OVERVIEW

Platform for building AI automation agents for university staff. Enables non-technical users to create workflows for document checking, email responses, and other automated tasks without writing code.

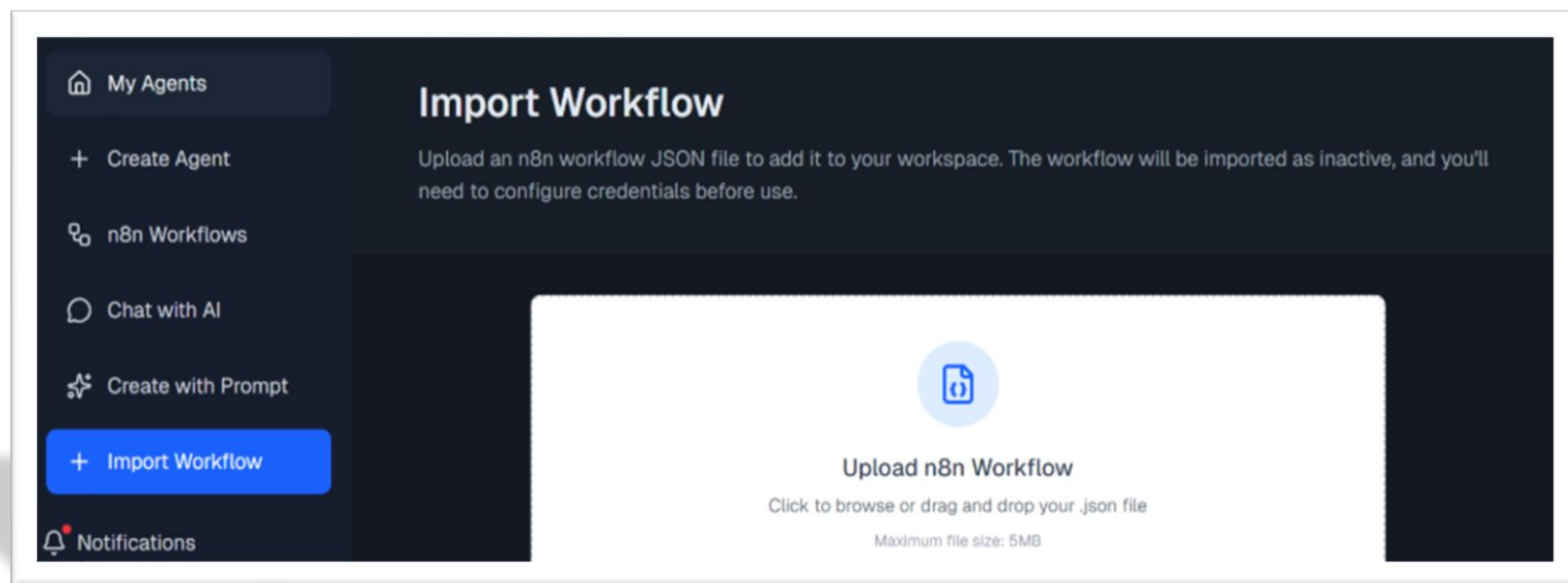
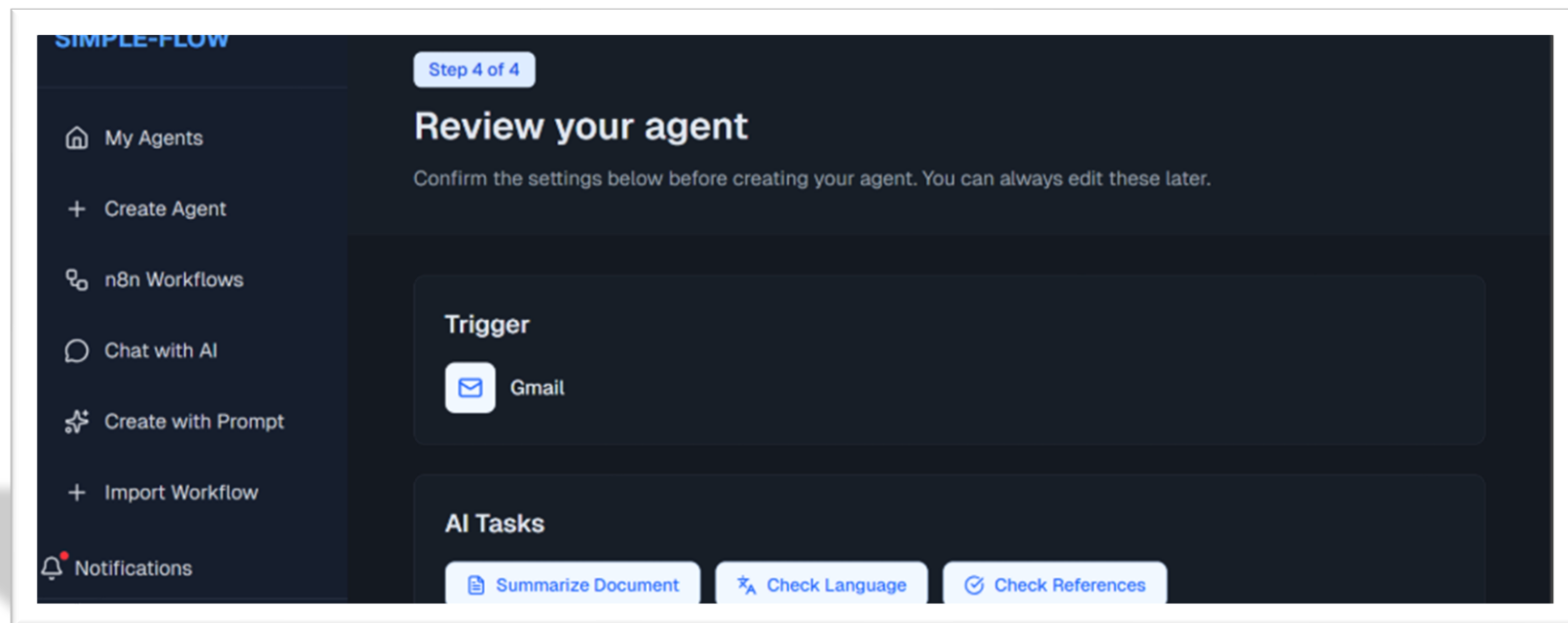
TECHNICAL IMPLEMENTATION

4-step workflow builder on n8n. Frontend: Next.js, React, TypeScript, Tailwind CSS. Backend: FastAPI, PostgreSQL, Prisma. Infrastructure: Docker and Kubernetes managing scalable n8n instances per user with OpenAI API integration.

SKILLS

Next.js · React · TypeScript · FastAPI · PostgreSQL · Docker · Kubernetes · n8n · OpenAI API

Role: Communication Manager · Staff communication, n8n workflow builder, builder, import function and frontend



Machine Vision

YOLO

YOLO Object Detection in Synthetic Environments

Solo Project

PROJECT OVERVIEW

Deep learning object detection system using YOLO to detect vehicles and pedestrians in a synthetic urban environment generated from Grand Theft Auto V — providing realistic training data without real-world data collection.

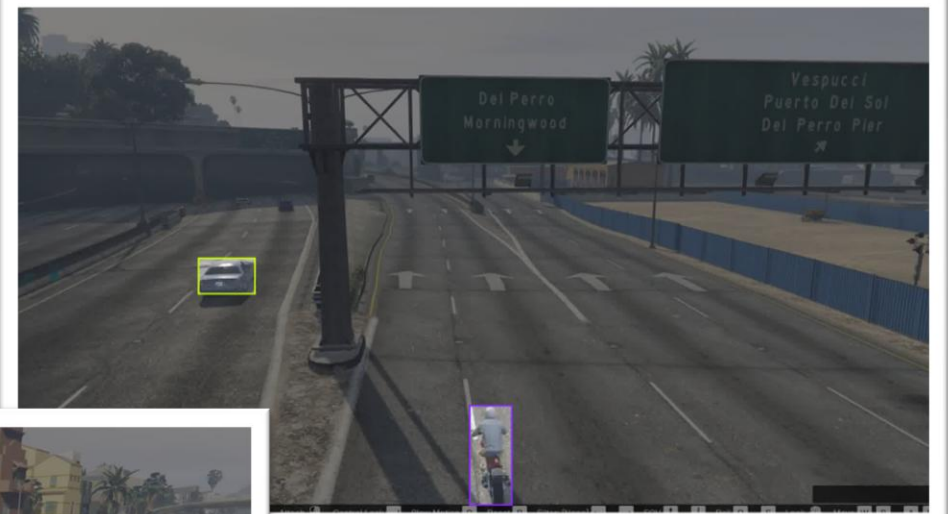
DATASET ENGINEERING

Image format conversion, annotation processing, and transformation of 3D simulation metadata into 2D bounding boxes. Full pipeline from raw game data to YOLO-compatible training set. Trained and evaluated in Google Colab.

SKILLS

Computer Vision · YOLO · Object Detection · Dataset Engineering · Synthetic Data · Python · Deep Learning

Role: Solo Project · Full pipeline from dataset preparation to model training and training and evaluation



MariLab

Simulator Data Analysis & AI Feedback

RoboAI Academy · Merilogistiikan Tutkimuskeskus

WHAT I BUILT

Extended a real-time tugboat simulator monitoring tool into a full training analysis platform. Built end to end: local LLM integration via Ollama for offline AI coaching feedback, behavioral analytics engine, GPS trajectory heatmaps, and side-by-side run comparison.

KEY TECHNICAL DECISIONS

Chose Ollama for offline LLM operation (no cloud dependency on restricted server). Used Plotly over Mapbox for GPS heatmaps. Fixed a CO₂ calculation bug computing voyage totals from only the last 60 samples instead of full run history.

STACK

Ollama (gemma4) · FastAPI · PostgreSQL · Alembic · Plotly · Podman / Podman / systemd

Role: AI Engineer · LLM integration, behavioral analytics, heatmaps, run comparison, Alembic migrations, bug fixes

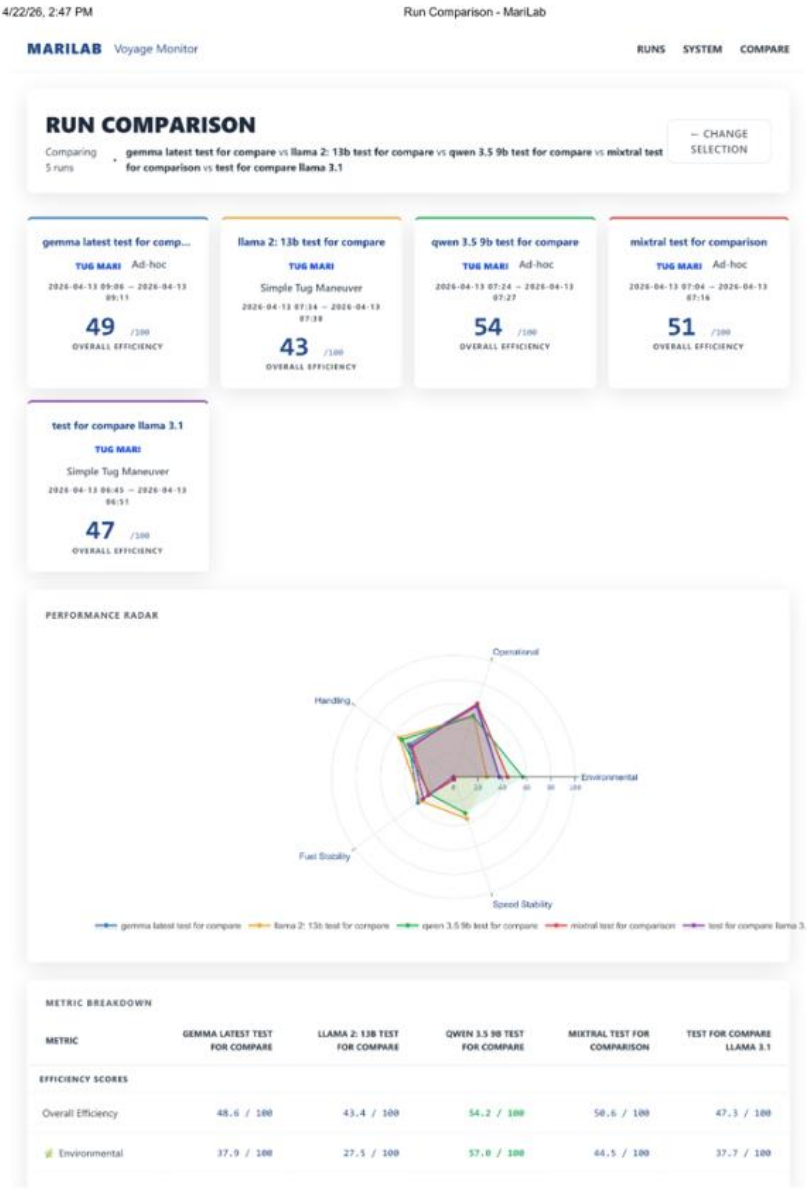


VOYAGE ANALYSIS

TRAJECTORY → FUEL RATE (KG/H)

EMISSIONS OVER TIME

127.0.0.1:8000/runs/3cb44426-661c-44f4-bbac-50200c9942a9/feedback-view



Hackathons

🥈 2nd Place

Robotics / Mechanical + Automation

Awarded 2nd place in a multidisciplinary hackathon focused on industrial automation. Designed a gripper using a vacuum-based mechanism for reliably picking metal cylinders from an unsorted container. Emphasised mechanical simplicity and robust object handling in industrial environments.

Mechanical Design

Robotics

Vacuum Gripping

Rapid Prototyping

Industrial Automation

🥉 3rd Place

AI / Computer Vision / Ethics

Awarded 3rd place in a hackathon on assistive technologies and ethical AI. Developed a proof-of-concept AI assistant using a depth camera for fall detection and elderly monitoring, with strong emphasis on privacy preservation by avoiding RGB imaging.

AI

Computer Vision

Depth Perception

Ethical AI

Assistive Technologies

Privacy Preservation